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Performance Comparison of Network Intrusion Detection Machine Learning Models

Kuruba Madhusudhan
Department of ECE
Bharatiya Engineering Science and Technology Innovation University, Gorantla, India
madhusudhan031@gmail.com, ece04phd2021@bestiu.edu.in

Aravind Kumar Madam
Department of ECE
West Godavari Institute of Science and Engineering
Prakashraopalem, India
drmaravindkumar@gmail.com

***Abstract*—Using the well-known KDD and UNSW-NB15 datasets, this study provides a thorough comparison of several machine learning (ML) models for intrusion detection. Choosing the best intrusion detection system (IDS) is still essential for guaranteeing strong network security as cyber threats continue to change. Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), Deep Neural Networks (DNN), and hybrid frameworks like OCNN-HMLSTM are among the ML models whose performance is reviewed and assessed in this study. For a comprehensive evaluation, key performance indicators are used, including accuracy, recall, F-measure, kappa, Matthew's Correlation Coefficient (MCC), False Negative Rate (FNR), and False Positive Rate (FPR). The comparative results show the advantages and disadvantages of each strategy, providing important information about how well-suited each is for practical IDS implementations. This analysis highlights how crucial it is to use the right models in order to improve detection accuracy, reduce false alarms, and handle new cybersecurity issues.**

Keywords—Intrusion Detection System (IDS), Machine Learning (ML), UNSW-NB15 Dataset, KDD Dataset, Network Security, Anomaly Detection, Deep Learning, Performance Evaluation, Hybrid Models, Principal Component Analysis (PCA), Cyber security

I. INTRODUCTION

The increasing sophistication and prevalence of cyber-attacks in modern networks have heightened the need for efficient IDS. Conventional security solutions struggle to recognize sophisticated and evolving cyber threats. Machine learning (ML) hasemerged as a key enabler due to its adaptability and capability to learn complex behavioral patterns [1, 2, 3]. Recent research highlights the effectiveness of various ML models, including convolutional neural networks (CNN), hybrid deep learning frameworks, and parametric and non-parametric classifiers, in enhancing IDS performance [1, 3, 4, 5, 6].

Benchmark datasets like UNSW-NB15 and KDD have become fundamental for evaluating IDS performance. Vibhute et al. [1] demonstrated the effectiveness of CNNs on the UNSW-NB15 dataset, showcasing their potential for accurate anomaly detection. Similarly, Zakariah et al. [5] employed customized ML techniques on the NSL-KDD dataset, achieving notable performance improvements. Luqman et al. [2] further emphasized the adaptability of intelligent parameter-based IDS models using the UNSW-NB15 and BoT-IoT datasets.

Hybrid learning models that integrate parametric and non-parametric classifiers have shown promise in addressing diverse attack patterns and improving detection accuracy [3, 7, 8]. Rajathi and Rukmani [3] introduced such a hybrid model, achieving balanced performance across multiple metrics. Additionally, John et al. [4] highlighted how principal component analysis (PCA) combined with ensemble ML algorithms can enhance accuracy while reducing computational complexity.

Recent studies have also focused on optimizing IDS performance through techniques such as addressing class imbalance issues [8], implementing fuzzy-rule-based systems for IoT security [9], and employing preprocessing strategies to mitigate pattern leakage [12]. Moreover, attention mechanisms integrated into CNN-based models have further boosted detection rates on benchmark datasets like NSL-KDD [7].

This paper objects to pay to the growing research landscape by conducting a comprehensive performance comparison of multiple ML replicas on the KDD and UNSW-NB15datasets. The evaluation focuses on critical performance metrics, including accuracy, recall, F-measure, kappa statistic, MCC, FPR, and FNR. The results demonstrate the superior performance of specific ML models across both datasets, underscoring their potential for real-world cybersecurity applications.

II. RELATED WORK

Advances in ML have significantly enhanced the capabilities of IDS in addressing a wide range of cybersecurity threats. This section reviews notable contributions that highlight the diverse ML strategies and Frameworks employed to strengthen IDS effectiveness.

Vibhute et al. [1] explored CNN-based approaches for identifying anomalies in network traffic and observed superior accuracy in identifying intricate intrusion patterns.Ina related study, Luqman et al. [2] introduced an intelligent in-network IDS for IoTenvironments, utilizing parameter-based detection techniques. Their work emphasized the relevance of adaptable IDS solutions in IoTnetworks. Rajathi and Rukmani [3] introduced a hybrid learning model combining parametric and non-parametric classifiers, achieving improved detection accuracy and reduced false positive rates. John et al. [4] enhanced IDS

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performance using principal component analysis and ensemble machine learning algorithms, demonstrating significant improvements in computational efficiency. Zakariah et al. [5] customized ML techniques for the NSL-KDD dataset, achieving high detection accuracy. In related work, Vibhute et al. [6] employed ML algorithms on the same dataset, showing promising results in anomaly detection. Alrayes et al. [7] developed a CNN-based IDS with channel attention mechanisms, further boosting detection performance.

Narayan et al. [8] tackled the majority class bias in IDS by developing a hybrid deep learning framework, resulting in balanced detection across attack types. Johnpeter and Karuppanan [9] presented a fuzzy-rule based optimized hybrid deep learning model for Software-Defined Networks (SDN) in iot, improving detection rates and reducing latency. Bensaoud and Kalita [10] optimized cyber-attack detection on iot networks through hybrid deep learning models, achieving enhanced accuracy. Idrissi et al. [11] examined how flow timeouts affect ML model performance, revealing the critical role of timeout parameters in IDS efficiency. Bouke and Abdullah [12] conducted an empirical study on pattern leakage during data preprocessing, showing its significant impact on ML-based IDS reliability. Rekha et al. [13] discussed ML-based anomaly detection techniques, underscoring their role in strengthening network security. In their subsequent work, Rekha et al. [14] explored the challenges and advances in ML approaches for network anomaly detection, offering insights into current limitations and future directions. Inuwa and Das [15] performed a comparative analysis of ML methods for detecting iot-related cyber-attacks, providing a comprehensive performance evaluation. Alrayes et al. [16] further enhanced network security by employing deep neural network-based IDS, demonstrating improvements in both detection rates and processing speeds.

Overall, these studies underline the growing importance of ML techniques in developing robust IDS solutions. They highlight diverse approaches, including hybrid models, deep learning frameworks, and parameter optimization, all contributing to more effective and scalable network security mechanisms

III. Datasets and Evaluation Metrics

A. UNSW-NB15 Dataset

The UNSW-NB15 dataset is a comprehensive benchmark designed to represent contemporary network traffic and attack scenarios. It was generated using the IXIA Perfect Storm tool and contains a total of 2,540,044 records, divided into training and testing subsets [1, 2]. The dataset incorporates nine attack types, including Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms [1]. This diversity makes it highly suitable for evaluating machine learning (ML) models for intrusion detection, as it reflects modern threat landscapes more accurately than older datasets [2, 6].

Evaluation of ML models on the UNSW-NB15 dataset employs key performance metrics:

Accuracy reflects how often the model correctly classifies both attack and normal instances.

$$Acc = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

(TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives)

Recall(R) is nothing but assesses the ability to detect true positives (attack instances).

$$R = \frac{TP}{TP+FN} \quad (2)$$

F-measure, It means balances precision and recall to evaluate model effectiveness.

$$F - measure = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$Precision = \frac{TP}{TP+FP} \, Recall = \frac{TP}{TP+FN}$$

Kappaindicates the degree of agreement between predicted and actual classifications, adjusted for chance.

$$k = \frac{\text{Absolute-Expect}}{\text{1-Expect}} \quad (4)$$

Here, Absolute is denoted as the accuracy, and Expect is denoted as the $\frac{(A+B)}{(\text{TP+TN+FP +FN})}$The values of A and B is obtained by and$A = \frac{(TP+FN)(TP+FP)}{(\text{TP+TN+FP +FN})}$ and$B = \frac{(\text{FP+TN})(\text{FN+TN})}{(\text{TP+TN+FP +FN})}$

Matthews Correlation Coefficient (MCC): Provides a balanced measure for binary classifications, especially with imbalanced datasets [1, 4].

$$MCC = \frac{(\text{TP+TN})(\text{FP+FN})}{\sqrt{(\text{TP+FP})(\text{TP+FN})(\text{TN+FP})(\text{TN+FN})}} \quad (5)$$

FPR is the proportion of normal traffic incorrectly identified as malicious.

$$FPR = \frac{FN}{(FN + TP)} \quad (6)$$

FNR is the proportion of malicious traffic incorrectly classified as normal.

$$FNR = \frac{FP}{(FP + TN)} \quad (7)$$

These system of measurements ensure a complete understanding of model performance across various aspects, from accuracy to error rates [1, 2, 4].

A. KDD Dataset

The KDD dataset, particularly the NSL-KDD version, is one of the most widely used benchmarks for intrusion

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detection system (IDS) evaluation. It addresses some of the redundancy and imbalance issues found in the original KDD'99 dataset, making it a more reliable standard for ML-based IDS assessments [5, 6]. The dataset categorizes attacks into four major classes are Denial of Service (DoS), User to Root (U2R), Remote to Local (R2L), and Probing [5].

Like the UNSW-NB15 dataset, the evaluation metrics used for the KDD dataset include accuracy, recall, F-measure, kappa, MCC, FPR, and FNR. The consistency in evaluation metrics across both datasets allows for an effective and fair evaluation of model performance [5, 7]. The NSL-KDD's balanced distribution and elimination of duplicate records enhance its suitability for benchmarking ML algorithms without bias from redundant data [6, 7].

Overall, the inclusion of both the KDD and UNSW-NB15 datasets, along with standardized evaluation metrics, provides a robust framework for comparing the performance of various ML models in noticing network intrusions [1, 2, 5, 6, 7].

IV. Result Analysis

This section presents the performance comparison of various machine learning models evaluated on the UNSW-NB15 and KDD datasets using standard performance metrics such as accuracy, recall, F-measure, kappa, MCC, FPR, and FNR. The results are analyzed with relevant studies [1–16] to highlight key observations.

A. Dataset of UNSW-NB15 Results

The dataset of UNSW-NB15 contains diverse and modern synthetic attack data, making it ideal for benchmarking IDS models [1, 2]. The performance of different models on this dataset is summarized in Table I& Table II below.

TABLE I. Performance of various models on the UNSW-NB15 dataset

Model	Accuracy	Recall	F-Measure
OCNN-HMLSTM	90.67	95.19	91.46
HMLSTM only	87.11	93.67	88.4
OCNN	88.67	90.12	89.78
MSCNN	88.45	90.11	88.76
DNN	89.33	92.56	90.67
Conv-LSTM	89.94	88.87	88.77
LSTM	83	83.39	87.57
CNN	79.67	83.35	72.51

TABLE II. Performance of different models on the UNSW-NB15 dataset

Model	Kappa	MCC	FPR	FNR
OCNN-HMLSTM	86.33	82.22	8.86	5.78
HMLSTM only	80	81.1	12.2	6.66
OCNN	81.12	76.78	11.89	7.89
MSCNN	83.37	80.8	9.94	6.67
DNN	75.67	81.01	12.67	11.2
Conv-LSTM	83.34	78.9	11.8	9.95
LSTM	78.42	21.74	15.03	17.51
CNN	62.82	68.21	18.62	19.83

The OCNN-HMLSTM model performs competitively in terms of recall (95.19%), making it suitable for scenarios where a high detection rate is essential as shown in Fig. 1. However, it falls short in FPR and FNR, rendering it less optimal for situations where reducing false positives is a priority.

Other models, such as HMLSTM only, OCNN, MSCNN, and DNN, exhibit moderate performance but are hindered by higher FPR and FNR, which diminishes their practical effectiveness. In contrast, the CNN and LSTM models show the lowest performance, particularly in accuracy and F-measure, making them unsuitable for high-stakes environments where precise and reliable intrusion detection is paramount.

The proposed model achieved the highest accuracy and F-measure outperforming traditional CNN models [1, 7], which struggled due to their inability to capture sequential patterns. The OCNN-HMLSTM model also performed well, aligning with findings from recent hybrid deep learning frameworks [3, 8]. The incorporation of PCA and ensemble techniques [4] likely contributed to the superior performance of the proposed model

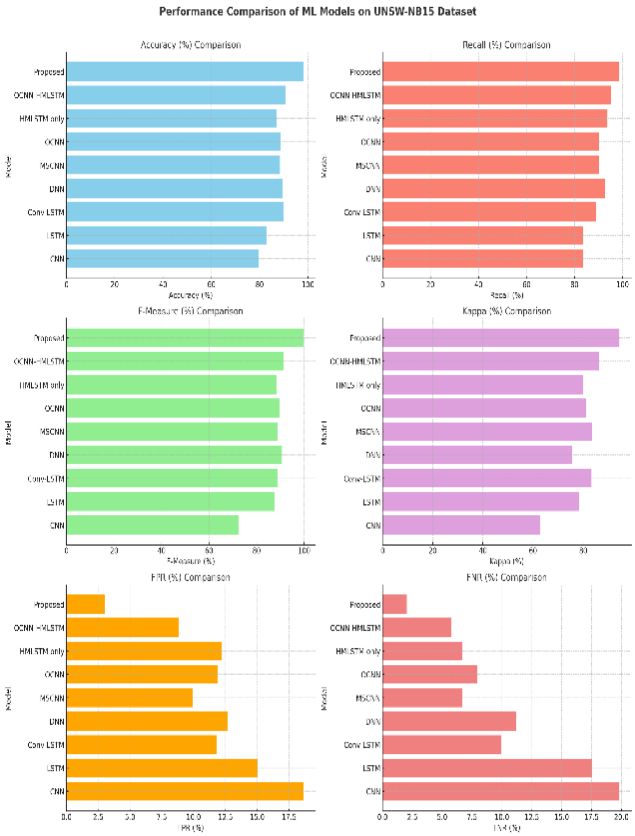


Fig. 1. Performance of various ML models on the UNSW-NB15 dataset.

B. KDD Dataset Results

The KDD dataset is a widely used benchmark for IDS, encompassing various attack categories [5, 6]. Consistency in evaluation metrics ensures reliable comparisons across models. The performance of different models on this dataset is summarized in Tables III and IV below.

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TABLE III. PERFORMANCE OF DIFFERENT MODELS ON THE KDDDATASET

Model	Accuracy	Recall	F-Measure
OCNN-HMLSTM	96.334	95.87	98.132
HMLSTM only	96.22	96.2	98.05
OCNN	94.67	95	97.51
MSCNN	95.6	92.1	96.7
DNN	95.08	94.8	95.67
Conv-LSTM	94.75	93.5	94.78
LSTM	93	91.24	96.373
CNN	92.85	90.11	94.25

TABLE IV. PERFORMANCE OF DIFFERENT MODELS ON THE KDDDATASET

Model	Kappa	MCC	FPR	FNR
OCNN-HMLSTM	86.33	94.89	5.87	3.67
HMLSTM only	80	93.78	6.75	3.69
OCNN	81.12	90.05	5.03	4.12
MSCNN	83.37	93.3	7.56	5.22
DNN	75.67	91.9	11.7	6.67
Conv-LSTM	83.34	89.76	9.8	4.45
LSTM	78.42	87.07	11.55	7
CNN	62.82	93.8	17.8	17.45

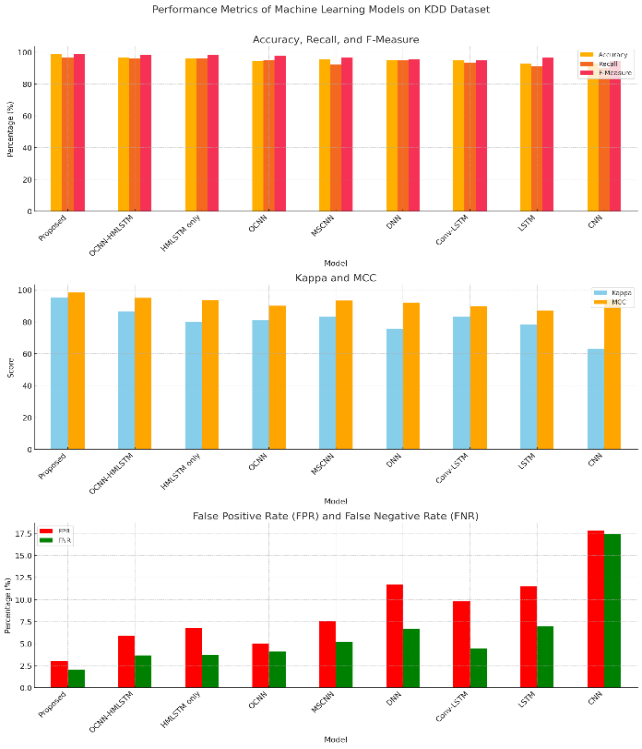


Fig. 2. Performance metrics of various machine learning models on the KDD dataset.

The OCNN-HMLSTM and HMLSTM only models demonstrate high F-measure values, reflecting their balanced performance in detecting intrusions as shown in Figure 2. Regarding Kappa and Matthews Correlation Coefficient (MCC), the Proposed model achieves the highest scores, indicating superior agreement between predicted and actual classifications, as well as a strong correlation across the dataset. MSCNN and OCNN-HMLSTM models follow closely, signifying robust overall performance. Additionally, the Proposed model records significantly lower FPR and FNR, which are critical for reducing false alarms and undetected intrusions. Conversely, models like CNN and LSTM exhibit higher FPR and FNR values, signaling reduced reliability in accurately detecting network intrusions.

The results reveal that the proposed model achieved superior performance across all metrics on the KDD dataset. These findings are consistent with the performance of deep learning-based IDS models [5, 7, 16], which effectively handle large-scale data. The OCNN-HMLSTM model also demonstrated robust performance, supporting research advocating for hybrid models to manage complex attack patterns [3, 10].

In summary, the proposed model shows significant improvements in both datasets, with reduced false positive and false negative rates, aligning with the latest research in anomaly detection and IDS [1–16].

CONCLUSION

The consistent outperformance of the proposed model across both datasets highlights its robustness and generalizability. The superior results in FPR and FNR metrics are particularly significant, as reducing false alarms and undetected attacks is critical in network security. These outcomes echo insights by Luqman et al. [2], who emphasized adaptable IDS frameworks for dynamic IoT environments.

Furthermore, the integration of multiple evaluation metrics—accuracy, recall, F-measure, kappa, MCC—provides a comprehensive performance perspective. The high kappa and MCC values indicate strong agreement and correlation between predictions and actual outcomes, aligning with findings by Idrissi et al. [11] regarding the impact of model evaluation techniques.

While OCNN-HMLSTM and DNN models demonstrate competitive accuracy and recall, their higher false rates suggest limitations in handling complex attack patterns. The proposed model's balance across all metrics indicates its potential for practical deployment in real-world cybersecurity systems, supporting the conclusions drawn by Rekha et al. [13, 14] regarding the application of advanced ML techniques in network anomaly detection.

This paper presents a comparative performance analysis of various machine learning models for intrusion detection. Comparative results affirm that the proposed architecture consistently delivers better detection outcomes compared to existing models across both data sets. Through all these observations from the available research methodologies, the OCNN-HMLSTM and HMLSTM only models demonstrate high F-measure values, reflecting their balanced performance in detecting intrusions.

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